Key things.

$$\phi(n) = \text{no. of } j,\ 1 \le j \le n,\ \gcd(j,n)=1$$
$$= |U(n)|$$

Given the prime factorization $n = \prod_{i=1}^{r} p_i^{a_i}$

then $\phi(n) = \prod_{i=1}^{r} p_i^{a_i - 1}(p_i - 1)$

Theorems.

Euler's Theorem

For $a \in U(n)$ $\quad a^{\phi(n)} \equiv 1 \pmod{n}$

special case for prime modulus

Fermat's Little Theorem

If $p \nmid a$ then $a^{p-1} \equiv 1 \pmod{p}$

Q1 For instance

$$\phi(27) = \phi(3^3) = 3^2(3-1) = 18$$

Q4 $\mathbb{Z}_{15}^{\times} = U(15) = \{ x \in \mathbb{Z}_{15} : \gcd(x,15)=1 \}$

multiplicative group.

$= \{1, 2, 4, 7, 8, 11, 13, 14\}$

for $a, b \in U(15)$ can we simplify

$\phi(15) = 8$

$a^{24} b^{15}$

$= (a^8)^3 \, b^{8+7}$

$= 1^3 \, b^8 \, b^7$

$= b^7 = b^{-1}$

Since Eulers theorem says $a^{\phi(15)} \equiv a^8 \equiv 1 \pmod{15}$

$\Rightarrow b^8 \equiv 1 \pmod{15}$

Q5. In $U(42)$, $\phi(42) = \phi(2 \cdot 3 \cdot 7)$
$= 12$

$a^{25} b^{23} = a b^{-1}$

Q6. 3^{100} has unit digit ____

Given any integer x, its unit digit is r, where $0 \le r \le 9$

and $x \equiv r \pmod{10}$

and $\phi(10) = 4$.

So $3^4 \equiv 1 \pmod{10}$

$3^4 = 81 \equiv 1 \pmod{10}$.

$\Rightarrow \quad 3^{100} \equiv \underline{3^0 \equiv 1}. \pmod{10}$

$3^{100} = (3^4)^{25}$

Chap 4 Cyclic Groups ~~AA~~

<u>Q23</u> Let G be a group and let $a, b \in G$

1. Claim $\boxed{|a| = |a^{-1}|}$

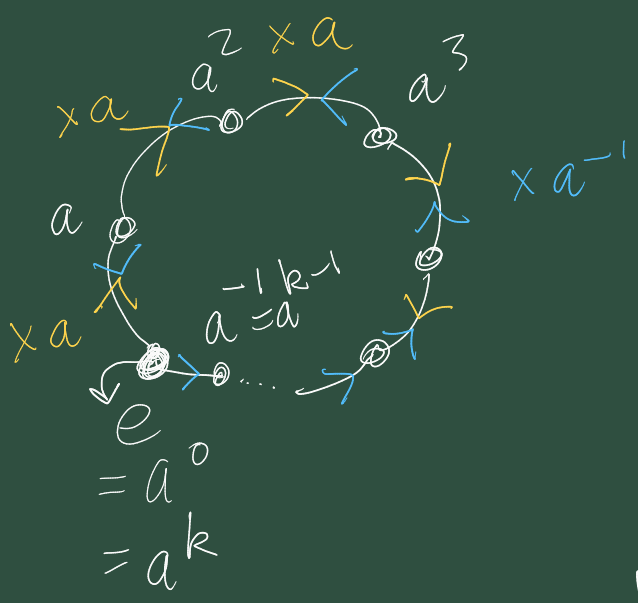
Remember

$|a| :=$ least positive integer $k > 0$ such that $a^k = e$.

$= |\langle a \rangle|$

<u>Proof</u> First with a picture

Suppose $|a| = k \in \mathbb{Z}$.



Picture of $\langle a \rangle$ and $\langle a^{-1} \rangle$

More formal algebraic proof.

Suppose $a^k = e$.

$\iff a^{k-1} = a^{-1}$ maybe...

$$a^k = e$$

$$\iff a^k a^{-k} = e a^{-k}.$$

$$\iff a^{k-k} = a^{-k}$$

$$\iff a^0 = a^{-k}$$

$$\iff e = a^{-k} = (a^{-1})^k$$

Hence $a^k = e \iff (a^{-1})^k = e.$

$$\implies |a| = |a^{-1}|$$

conjugation of a by g

2. Claim: $\forall g \in G$ $\boxed{|a| = |g^{-1}ag|}$

Continue 5:02.

~~So~~ A key fact that will help our subsequent calculations is

For any integer n

$$(g^{-1}ag)^n = g^{-1}agg^{-1}agg^{-1}ag \cdots g^{-1}ag$$

$$= g^{-1} a e a e a \cdots e a g$$

$$= g^{-1} a^n g$$

Suppose $\underline{a^k = e}$

$$(g^{-1} a g)^k = g^{-1} a^k g$$

$$= g^{-1} g$$

$$= e.$$

and in the other direction.

Suppose $(g^{-1} a g)^k = e$

$$\Rightarrow \quad g^{-1} a^k g = e$$

$$\Rightarrow \quad a^k = g e g^{-1} = g g^{-1} = e.$$

This shows

$$a^k = e \iff (g^{-1} a g)^k = e.$$

$$\Rightarrow \quad |a| = |g^{-1} a g|$$

3. Claim $|ab| = |ba|$

Suppose $(ab)^k = e$

$\iff a(ba)^{k-1} b = e.$

$\iff a^{-1} a (ba)^{k-1} b b^{-1} = a^{-1} e b^{-1} = a^{-1} b^{-1}$

$\iff (ba)^{k-1} = a^{-1} b^{-1}$

$\iff (ba)(ba)^{k-1} = ba\, a^{-1} b^{-1}$

$\iff (ba)^k = e.$

Therefore $|ab| = |ba|$

$(ab)^k = ab\,ab\,ab\,ab \ldots ab$

$(ba)^k = ba\,ba\,ba\,ba \ldots ba$

Chap 5

Q3

d) A product of non-disjoint cycles.

$\sigma = (17254)(1423)(154632)$

$\sigma \in S_7$.

Find the canonical form of σ, i.e. express σ as a product of disjoint cycles.

$$\sigma = (3)(5)(21467)$$

$$= (21467)$$

$$= (21)(14)(46)(67)$$

So σ is an even permutation as it takes 4 transpositions.

another version

$\sigma = (17)(72)(25)(54)(14)(42)(23)(15)(54)(46)(63)(32)$

So $\sigma \in A_7$.

<u>Q 12</u> Actually this generalises to the statement.

For any group G and element $g \in G$ with finite order k.

Claim $g^i = g^j \iff i \equiv j \pmod{k}$

"$\Longrightarrow$"

Suppose $g^i = g^j$

$\iff g^i g^{-j} = e$.

$\iff g^{i-j} = e$.

$\iff k \mid (i-j)$, since $|g| = k$

$\iff i \equiv j \pmod{k}$.